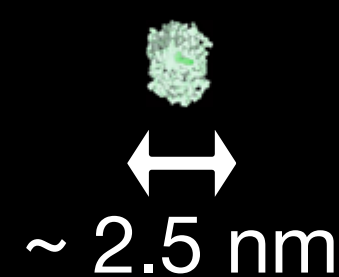


The Diffraction Limit

Fluorescent proteins frequently used in biological research are small

Green Fluorescent Protein (GFP)

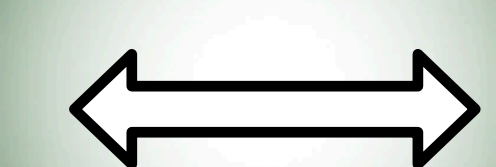


Even smaller than the wavelength of light they emit (~ 500 nm)

Interference of light results in a point-spread function (PSF) like the Airy disk

Structures smaller than the critical distance cannot be resolved by traditional light microscopes

$d \approx 200 \text{ nm}$

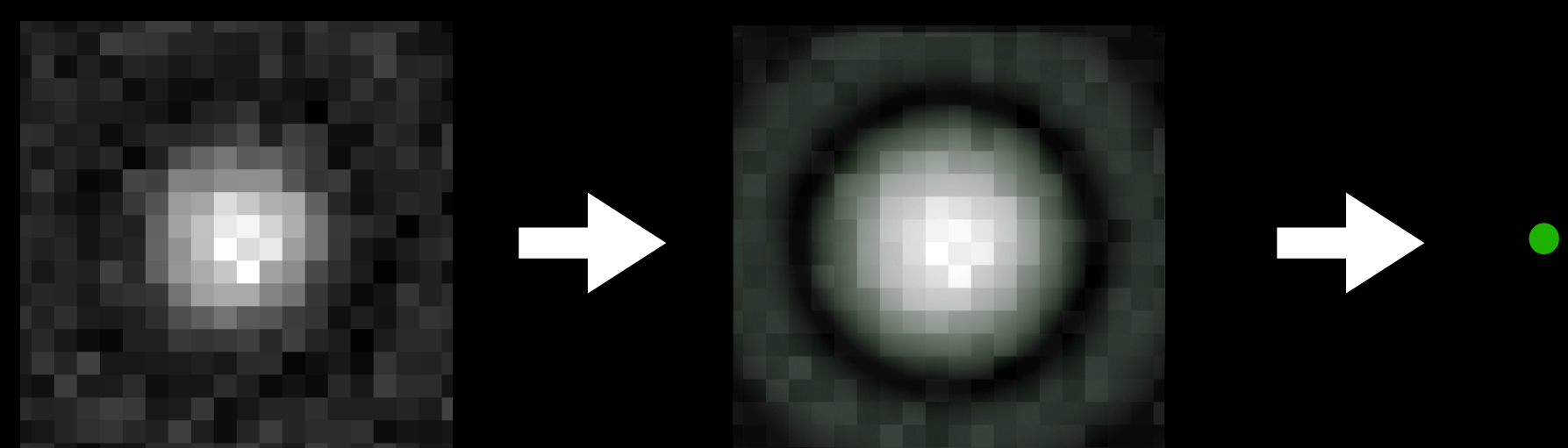


Rayleigh: $d \geq 0.61 \frac{\lambda}{NA}$ Abbe: $d \geq 0.5 \frac{\lambda}{NA}$

Super-Resolution Microscopy

Resolutions beyond the diffraction limit are possible

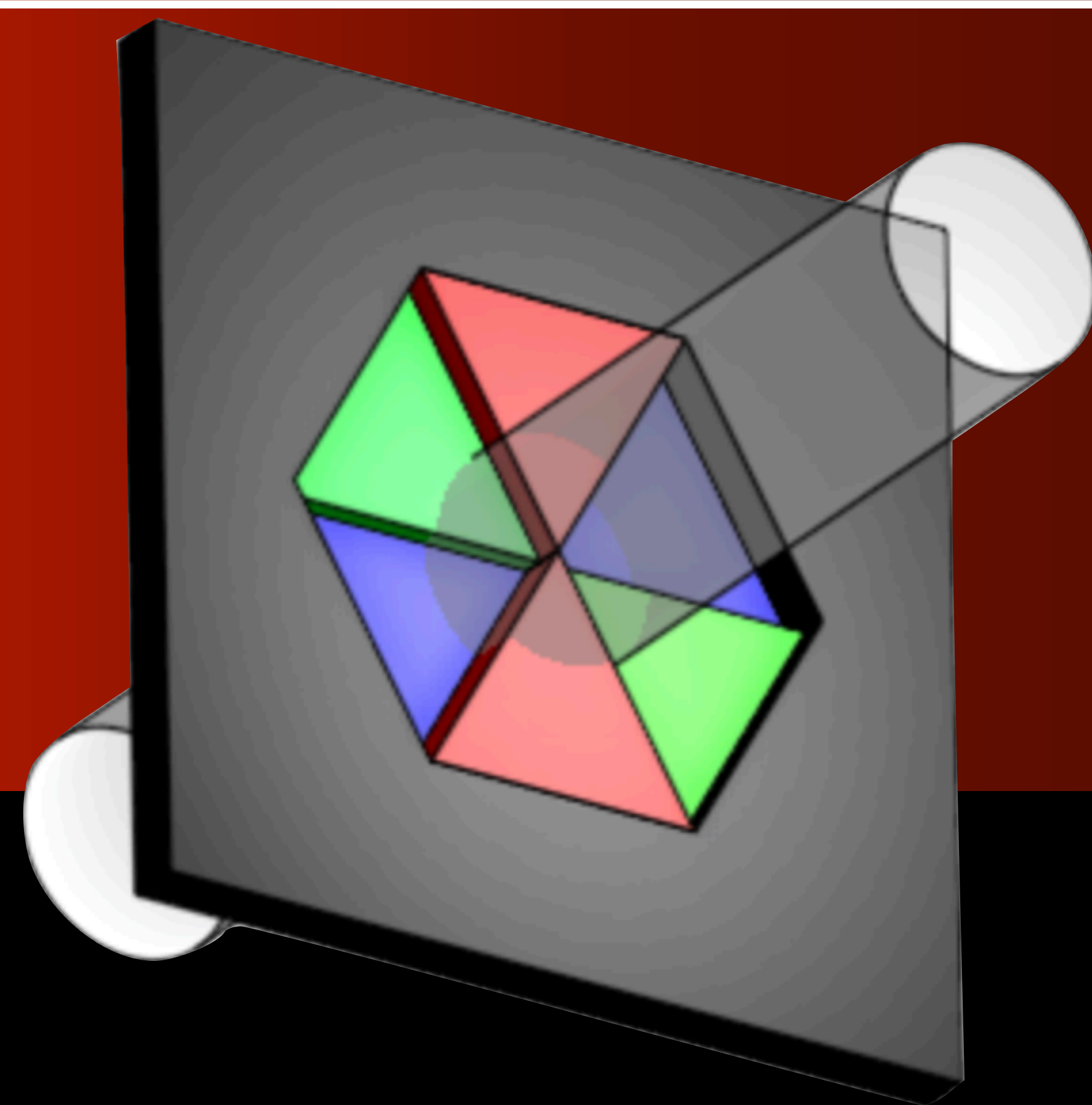
Measure Fit Reconstruct



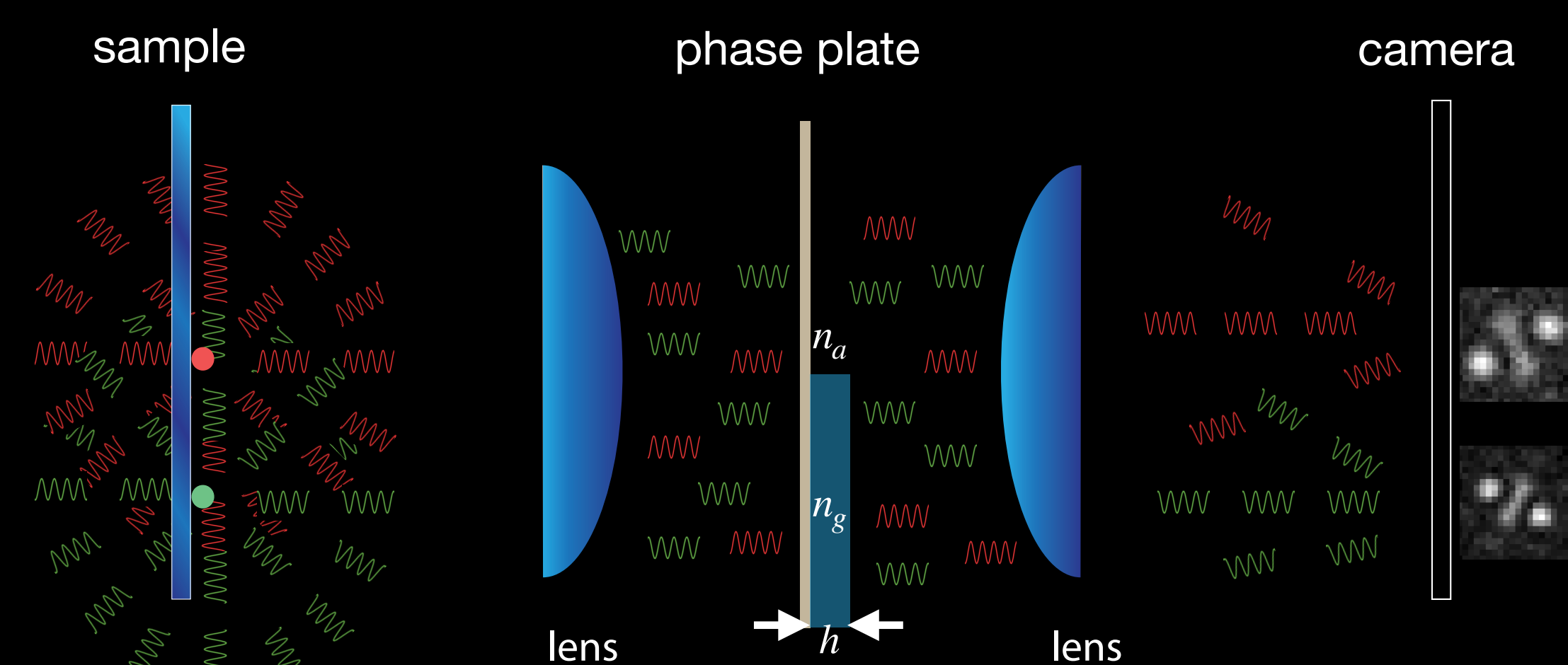
Single-molecule localization methods can achieve spatial resolutions up to the *Cramer-Rao Bound* (~10 nm), related to the Fisher Information

$$CRB_{\vec{\theta}}(\vec{\theta}) \leq \sqrt{F_{ii}^{-1}(\vec{\theta})}; \quad \vec{\theta} = [x_0, y_0, z_p, \lambda, \dots]$$

STORM, PALM, DNA-PAINT, and many other methods can resolve nanometer-scale structures



PSF Engineering with a Glass Phase Plate



The phase plate imparts a wavelength-dependent phase shift to the emission wavefront to make color-coded PSFs

$$\Delta\phi(\lambda) = \frac{2\pi}{\lambda}(n_g - n_a)h$$

Chernoff Information

The best way to classify an observation

$\nu =$ as either

$\nu_1 =$ or $\nu_2 =$ or $\nu_3 =$

is to compare the Chernoff Information (or Chernoff Bound) it has with each known class. This quantity tightly bounds the probability of a misclassification error (Bayes' error)

$$C(\beta_i) = \nu_i \frac{(\beta_i - 1)(\log(\frac{\beta_i - 1}{\log \beta_i}) - 1) + \log \beta_i}{\log \beta_i}; \quad \beta_i = \frac{\nu}{\nu_i}$$

The class that provides the tightest bound is the best classification decision we can make

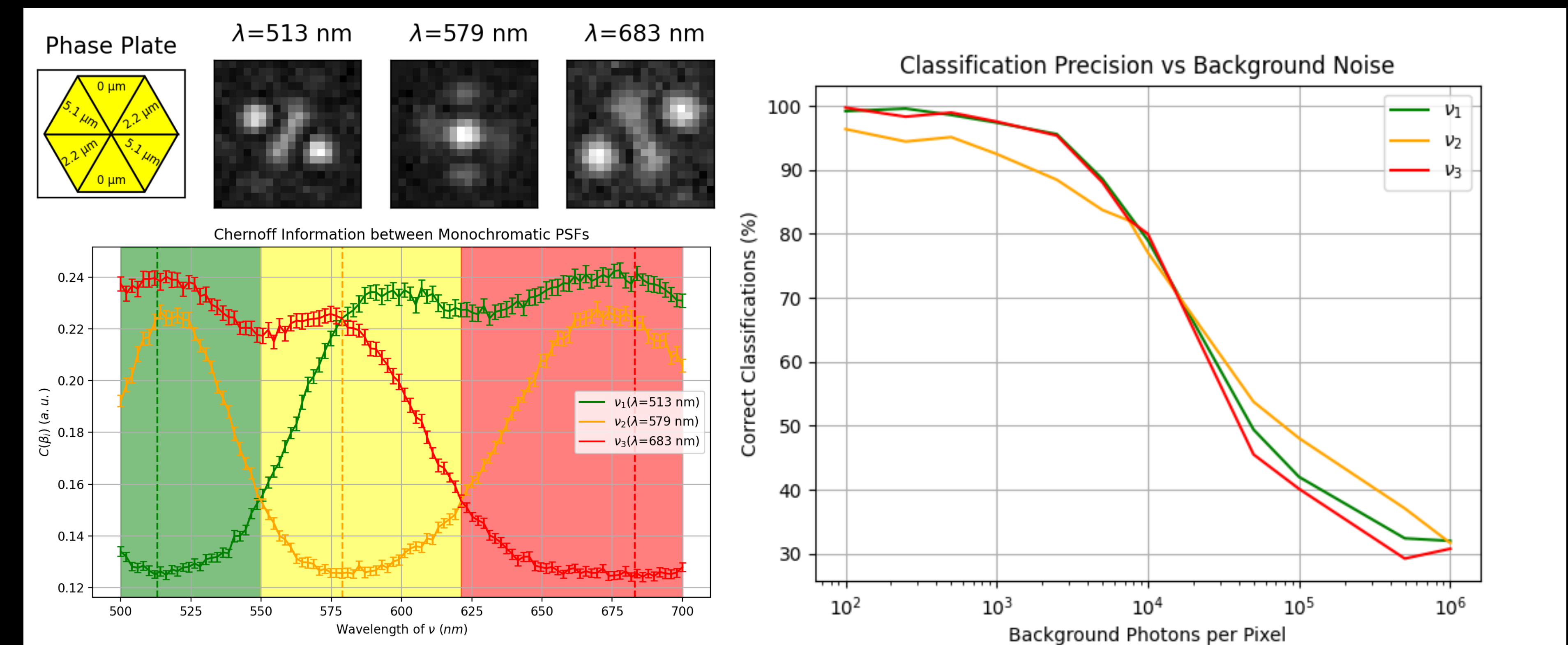
If $C(\beta_1) < C(\beta_2)$ and $C(\beta_1) < C(\beta_3)$, then $\nu \rightarrow \nu_1$

More Chernoff Information between classes yields more distinguishable PSF patterns

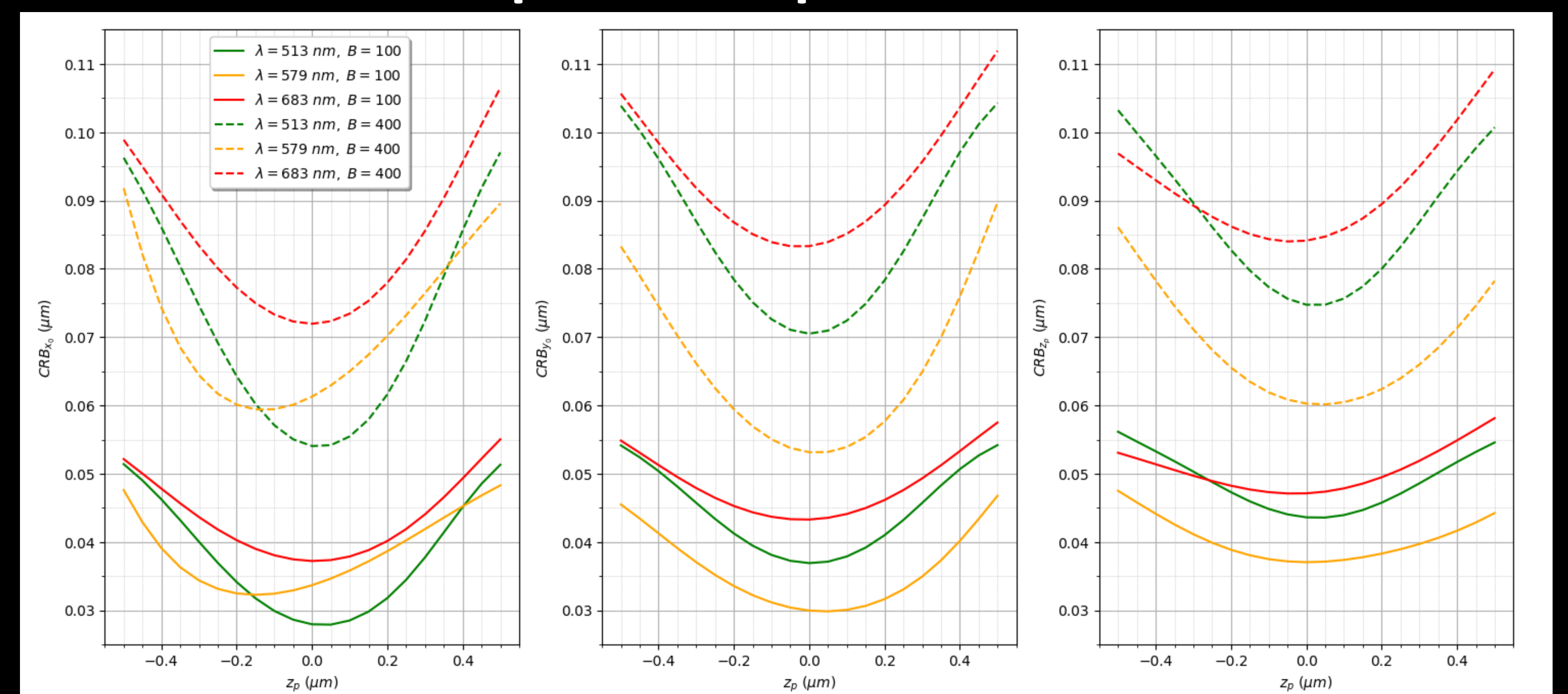
A Hexagonal Phase Plate for Optimal 3-Color Super-Resolution Microscopy

Matthew Hogan, Dept. of Physics & Astronomy
Jordan Gerton, Dept. of Physics & Astronomy
Erik Jorgensen, School of Biological Sciences
In collaboration with Brian Baker and Utah NanoFab

Improved Fluorophore Classification...

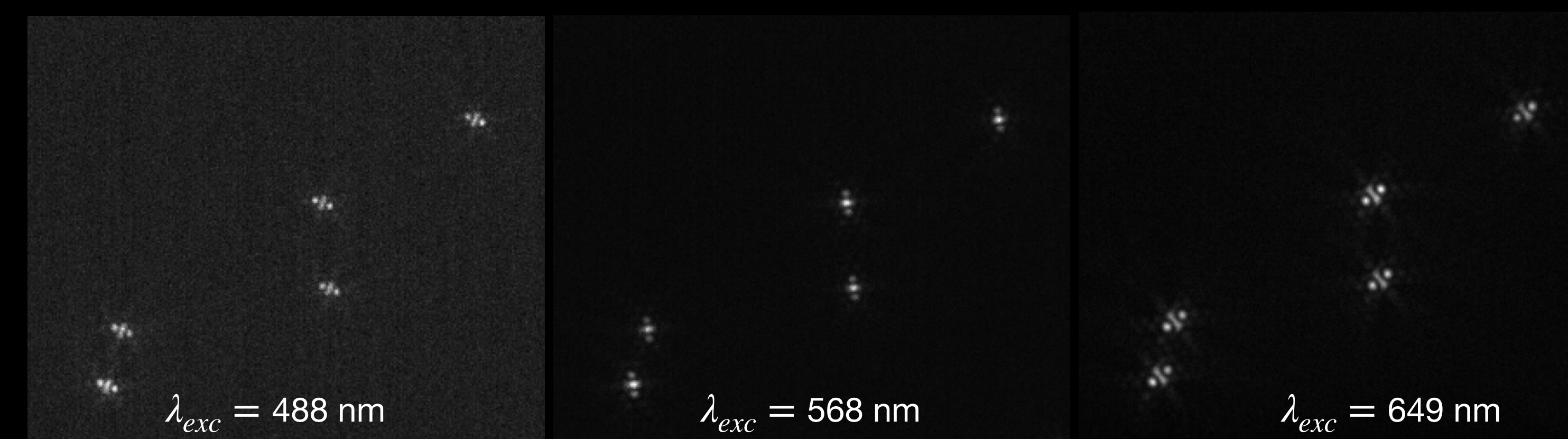


... at the Expense of Spatial Resolution



Results above describe the best-case precisions for simulated Hex-PSFs of 2500 photons

Diffraction-Limited Fluorescent Microspheres



Sequential laser excitations of a ~400 μm² field of view of 0.1 μm Tetraspeck™ microspheres. Each microsphere has green (513 nm), yellow (579 nm), and red (683 nm) fluorescent molecules attached

References

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- F. Nielsen, "An Information-Geometric Characterization of Chernoff Information," *IEEE Signal Processing Letters*, vol. 20, no. 3, pp. 269-272 (2013)
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Get in touch with me at
matthew.hogan@utah.edu
or check out my website
mhogan17.github.io

